

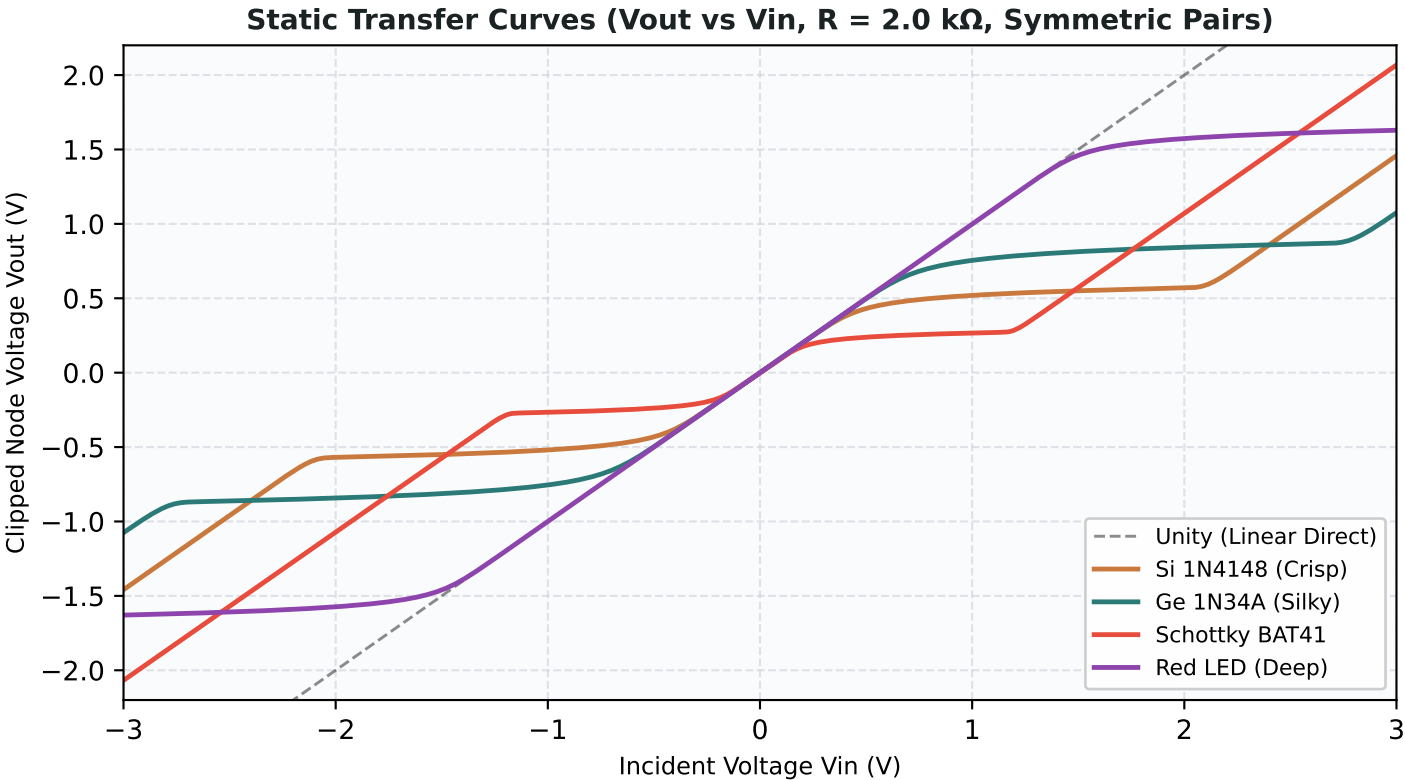
FRIED TOFU — MATHEMATICAL FOUNDATIONS

Wave Digital Filter (WDF) Diode Clipper Nonlinear Modeling & Physical Acoustics

Kijjaz Audio DSP

1. Circuit Model and Kirchhoff Wave Port Formulation
- An analog diode clipper driven through source/network resistance R obeys:
$$a = V + R \cdot ID(V)$$
where 'a' is the driving incident wave voltage, V is the node voltage across the diodes, and $ID(V)$ is the net semiconductor junction current governed by Shockley's physical equations:
$$ID(V) = I_{s,A} \cdot (\exp(V / (N_A \cdot V_t)) - 1) - I_{s,B} \cdot (\exp(-V / (N_B \cdot V_t)) - 1)$$
Here, $V_t = k_B \cdot T / q \approx 25.864 \text{ mV}$ is the thermal voltage at 300 K, I_s is the reverse saturation current, and N is the ideality emission factor.
2. Damped Newton–Raphson Solver & Guaranteed Convergence
- The node voltage V is found per-sample as the unique zero of the objective function:
$$F(V) = V + R \cdot ID(V) - a = 0$$
$$F'(V) = 1 + R \cdot [(I_{s,A} / (N_A \cdot V_t)) \cdot \exp(V / (N_A \cdot V_t)) + (I_{s,B} / (N_B \cdot V_t)) \cdot \exp(-V / (N_B \cdot V_t))]$$
Because $R > 0$ and incremental conductance $gd(V) > 0$ for all real V , $F'(V) \geq 1$ everywhere.
Hence $F(V)$ is strictly monotonically increasing with a single real root. Damped Newton iterations:
$$V_{\{k+1\}} = V_k - F(V_k) / F'(V_k)$$
with warm-start memory state $V_0 = V[n-1]$ converge quadratically within 3 to 8 iterations to $|\Delta V| < 10^{-9} \text{ V}$.

Diode Profile	I_s (A)	N	Knee	Physical Characteristics & Sonic Character
Si 1N4148 (Crisp)	2.52×10^{-9}	1.752	$\sim 0.62 \text{ V}$	Crisp silicon edge, sharp conduction, bright harmonic bite
Ge 1N34A (Silky)	2.00×10^{-10}	2.190	$\sim 0.28 \text{ V}$	Soft germanium knee, early saturation, warm silky compression
Schottky BAT41	2.00×10^{-8}	1.050	$\sim 0.32 \text{ V}$	Flash-fried low barrier, rapid current rise, aggressive fuzz
Red LED (Deep)	9.30×10^{-19}	1.840	$\sim 1.75 \text{ V}$	High forward headroom, open dynamic range, valve-like push



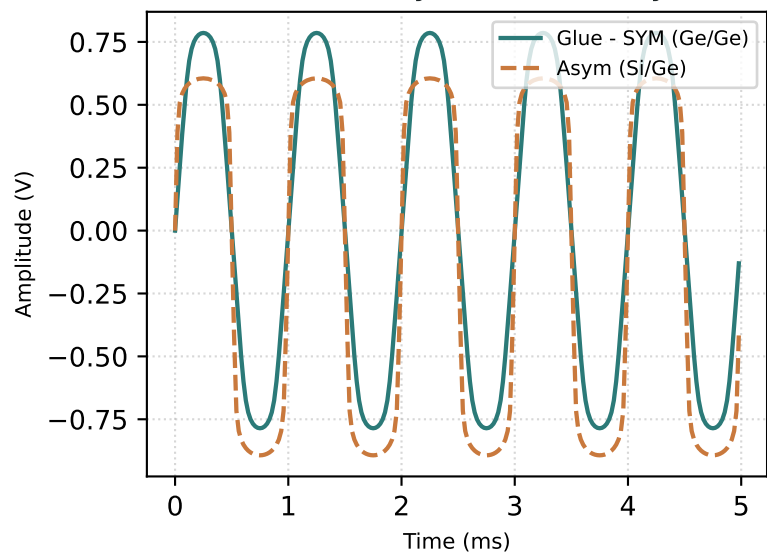
FRIED TOFU — DISTORTION & HARMONIC BENCHMARK

Test Tone: 1.0 kHz Sinusoid

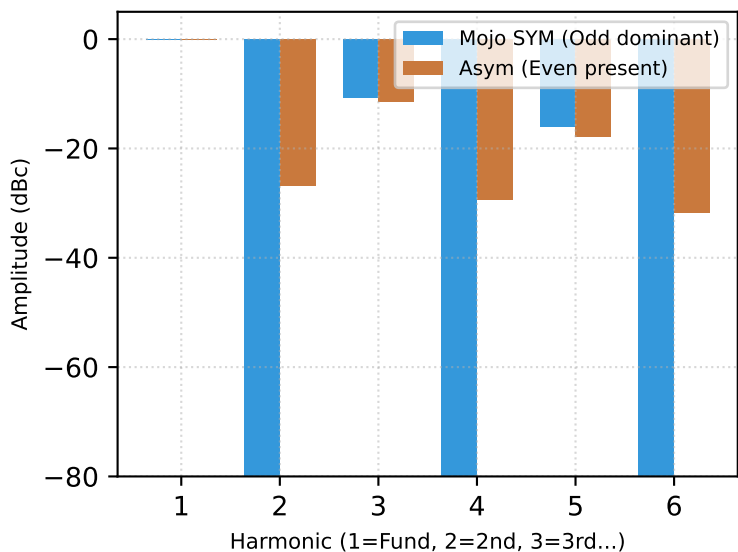
Empirical Measurements: Total Harmonic Distortion (THD) & Harmonic Spectra across 6 Presets

Preset Setting	Diode Configuration	Drive	R (Ω)	THD (%)	THD (dB)	H2 (dBc)	H3 (dBc)
Glue — SYM	1N34A / 1N34A	+6.0 dB	1000 Ω	8.43 %	-21.5 dB	-171.6 dB	-21.6 dB
Mojo — SYM	1N4148 / 1N4148	+22.0 dB	2000 Ω	34.72 %	-9.2 dB	-169.9 dB	-10.7 dB
Glue/Mojo — ASYM	1N4148 / 1N34A	+16.0 dB	1600 Ω	31.16 %	-10.1 dB	-26.8 dB	-11.4 dB
Germanium Fuzz	1N34A / 1N34A	+30.0 dB	630 Ω	36.52 %	-8.7 dB	-169.8 dB	-10.4 dB
Flash Crunch	BAT41 / BAT41	+18.0 dB	2200 Ω	33.88 %	-9.4 dB	-170.0 dB	-10.9 dB
Deep Open	LED / LED	+24.0 dB	3300 Ω	34.47 %	-9.3 dB	-169.9 dB	-10.6 dB

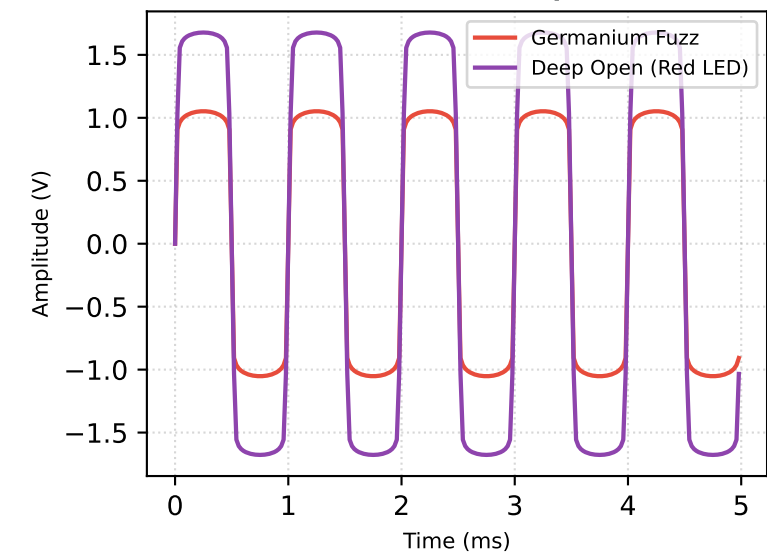
Waveforms: Symmetrical vs Asym



Harmonic Content: SYM vs ASYM (dBc)



Waveforms: Fuzz vs Open LED



FFT Spectrum: Asymmetric (Si + Ge)

